

Vatsal Hitesh Lodaya

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Profile

Motivated and detail-oriented B.Tech Computer Engineer with a strong foundation in programming, software development, and data structures. Experienced in building web applications and management systems using the MERN stack, and multi-agent and RAG systems with generative AI, with a growing interest in data science and analytics. Adept at problem-solving, team collaboration, and fast learning. Passionate about developing efficient digital solutions, analyzing data-driven insights, and continually expanding technical knowledge through hands-on projects.

Education

Shah & Anchor Kutchhi Engineering College	2022-2026
Bachelor of Technology (B.Tech) Computer Engineering & Honours in Data Science	CGPA: 8

Experience

Associate Software Developer Intern Nimap Infotech LLP	June 2024 - July 2024
Assisted in web application development and debugging while collaborating with the development team on various software projects, gaining practical experience in the software and web development lifecycle.	

Projects

AI-Based Fish Detection System

- Developed an AI-powered system using YOLO-based object detection to identify and classify fish species from underwater images and live video streams with high accuracy.
- Implemented a web application with user authentication, image upload, and real-time detection features for efficient marine data analysis.
- Technologies Used:** Python, Flask, YOLOv11, HTML, CSS, JavaScript, MySQL

Companio

- Full-stack app to find a verified person nearby for the same activity: Google OAuth and email sign-in, nearby discovery through an indexed PostGIS query, with map pins randomized to protect real locations.
- Live chat over Socket.IO, connection requests, block and report flows; 166 unit and 88 end-to-end tests, k6 load tests on 5,000 seeded users, Docker deployment.
- Technologies Used:** Next.js, NestJS, TypeScript, PostgreSQL, PostGIS, Prisma, Redis, Socket.IO, Docker

EDITH - Multi-Agent AI Orchestrator

- Orchestrator plans a request (e.g. a PDF into slides) and hands steps to specialist agents for RAG, documents, Excel, PowerPoint, browsing, research, email and calendar, running independent steps in parallel.
- Human approval pauses the graph before risky actions; conversations are checkpointed, failing agents are isolated, and every run is traced.
- Technologies Used:** Python, LangGraph, LangChain, Groq, FastAPI, React, ChromaDB

RAG Assistant

- Upload PDFs and chat with them; answers are grounded in the document text with page-level citations.
- Hybrid retrieval (embeddings + BM25, reciprocal rank fusion, cross-encoder reranking), background indexing with resume/cancel, and OCR for scanned pages.
- Technologies Used:** React, Vite, FastAPI, Python, ChromaDB, Gemini, OpenRouter

Trade Advisor

- Verdict-only advisor for Indian markets: each rule section (intraday, swing, long-term, options, hedge) runs as a parallel LangGraph agent; an LLM only explains the verdicts.
- Opportunity scanner, bar-by-bar backtester with no lookahead, option strike picking by delta, and a daily PDF morning brief.
- Technologies Used:** Python, LangGraph, Streamlit, yfinance, Pandas, Plotly

Customer Churn Analysis Dashboard [↗](#)

- Analyzed 10,000+ customer records using SQL and Python to identify key churn drivers and built an interactive Power BI dashboard, cutting manual reporting time by ~40%.
- **Technologies Used:** SQL, Python, Pandas, Power BI

GYM Management System [↗](#)

- Digital platform that manages memberships, tracks training schedules, guides workouts, and handles equipment purchases, streamlining all gym operations in one system.
- **Technologies Used:** MERN Stack

Loan Default Prediction Model

- Built and compared ML models (Logistic Regression, Random Forest) achieving 87% accuracy, and deployed the model as a Streamlit app for real-time predictions.
- **Technologies Used:** Python, Scikit-learn, Streamlit

Credit Card Statement Parser

- Streamlit app that reads credit card statement PDFs, detects the bank and extracts card holder, last four digits, billing period, due date and total due.
- **Technologies Used:** Python, Streamlit, pdfplumber, Regex

Skills

- **Technical Skills:** C/C++, Python, SQL, MERN Stack, TypeScript, Next.js, FastAPI, Flask, Gen AI & LLMs, LangGraph, LangChain, RAG
- **Other Software:** Power BI, Tableau, MS Tools, n8n, Docker, PostgreSQL, MySQL, Redis
- **Soft Skills:** Communication, Teamwork, Problem Solving, Detail Oriented, Initiative, Time Management, Adaptability, Multi-Tasking, Creativity
- **Strengths:** Rapid learning, public speaking, leadership under pressure

Certifications

- AWS Cloud Practitioner Essentials, AWS Training and Certification
- Paid Internship and Training Program in Web Development, AcmeGrade
- Deloitte Australia Data Analytics Job Simulation on Forage, July 2025